

ADITYA KUMAR

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EDUCATION

Indian Institute of Technology Kharagpur

Bachelor of Technology in Mechanical Engineering

2021 – 2025

CGPA: 7.64/10

EXPERIENCE

Software Developer

House of Amber Advisory Private Limited

May 2025 – Present

Mumbai, India – Onsite

- Designed and scaled high-performance backend services using **Python** and **FastAPI** for enterprise-grade applications
- Built optimized **100+** RESTful APIs with robust validation, error handling, and authentication for seamless integrations
- Deployed containerized production services using **Docker** on **AWS Kubernetes**, ensuring high availability and scalability
- Enhanced system reliability through structured logging, monitoring, and automated testing workflows across services

Software Developer Intern

Mahi Mahi Marketing Solution

Mar 2025 – Jun 2025

Bangalore, India – Remote

- Developed **Maven.ai**, an AI-powered marketing platform automating blog, image, and video content using Generative AI
- Optimized AI models and integrated detailed analytics dashboards to improve content quality and campaign effectiveness

Full Stack Developer Intern

Dhruva Capital

Feb 2025 – Apr 2025

Kolkata, India – Remote

- Built **SwanSathi.com**, a gold-loan management platform with authentication, loan processing, and real-time tracking
- Developed **DhruvaCapital.com** using **React.js** and **Node.js** with a modern, responsive, SEO-optimized UI
- Integrated backend services and optimized database performance using caching and load balancing strategies

PROJECTS

Distributed File Storage | Golang, P2P Networks, Distributed Systems | [Link](#)

2026

- Built a robust P2P file storage network in **Golang** with decentralized node discovery, data replication, and redundancy
- Architected pluggable **store abstraction** supporting multi-node consistency, fault-tolerant state management, and availability
- Designed gossip-based peer routing and discovery protocols for efficient mesh communication and network resilience

Zero-Knowledge Proof Authentication Service | Rust, gRPC, Applied Cryptography, Docker | [Link](#)

2026

- Implemented the **Chaum-Pedersen ZKP** protocol in **Rust** over 1024-bit RFC 5114 Diffie-Hellman groups securely
- Built a challenge-response **gRPC** service using **tonic/prost** for proof validation and secure token issuance
- Containerized with **Docker Compose** and rigorously unit-tested all cryptographic operations at **1024-bit** parameters

Portmon | Rust, Linux, Networking, CLI | [Link](#)

2025

- Built high-performance **Rust** CLI tool to inspect active TCP/UDP ports and map them to running Linux processes
- Implemented functionality comparable to `sudo ss -tulpn` by parsing system socket and process metadata efficiently
- Designed secure, privilege-aware execution exposing detailed port, PID, protocol, and listening state information

RESEARCH EXPERIENCE

IIT Kharagpur Interactive Map & Navigation System | Next.js, Geospatial Systems, Algorithms

2025

- Designed interactive campus-wide navigation system with route discovery and location search
- Implemented graph-based pathfinding algorithms to compute optimal routes across academic blocks, hostels, and facilities

ACHIEVEMENTS

- Secured rank within top **2%** in **JEE Advanced 2021**, among 1.6 lakh+ aspirants nationwide
- Achieved **1400+** rating on **Codeforces** and solved **500+** DSA problems across competitive programming platforms
- Built and deployed real-world applications spanning scalable APIs, AI-powered platforms, and Linux-based system utilities

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Rust, C, C++, Golang, Solidity, C#, HTML, CSS

Frameworks & Libraries: FastAPI, Next.js, React.js, Node.js, Express, Django, NextAuth.js, ShadCN

Databases & ORMs: PostgreSQL, MongoDB, MySQL, Redis, SQLite, Prisma, Sequelize

DevOps & Cloud: Linux, Docker, Kubernetes, AWS (S3, EC2, Lambda, RDS), Nginx, Git, YAML, CI/CD

Tools & IDEs: VS Code, PyCharm, Jupyter Notebook, Postman, Zed, Sublime Text